1 Solution — GIAM 1.1 Problem 2

1.1 Problem

Write the set $\mathbb{Z}$ of integers using a singly infinite listing.

1.2 Setup — What "Singly Infinite" Means

The book's earlier display of $\mathbb{Z}$ as

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$$

is *doubly infinite*: the ellipses extend in **both** directions. A *singly infinite* listing has just one ellipsis — it's a sequence indexed by the natural numbers $\mathbf{1, 2, 3, \ldots}$, with a definite first element and no "left end" to worry about.

1.3 The Listing

Alternate sign, growing in magnitude each round:

$$\mathbb{Z} = \{\, 0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ 4,\ -4,\ \ldots\, \}.$$

Equivalently, the listing is given by the function $f : \mathbb{N} \to \mathbb{Z}$

$$f(n) = \begin{cases} \ \ n/2 & \text{if } n \text{ is even,} \\ -(n-1)/2 & \text{if } n \text{ is odd.} \end{cases}$$

(Equivalently in one piece: $f(n) = (-1)^n \lfloor n/2 \rfloor$.)

The first few values:

n	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
$f(n)$	0	1	-1	2	-2	3	-3	4	-4

Table 1.1

1.4 Why It Works — Every Integer Appears Exactly Once

We must check f is a *bijection* from $\mathbb{N}$ to $\mathbb{Z}$: every integer appears at least once (surjectivity) and at most once (injectivity).

Surjective. Given any $k \in \mathbb{Z}$, we can solve for the n with $f(n) = k$:

- If $k > 0$: take $n = 2k$. Then n is even and $f(n) = n/2 = k$. ✓
- If $k \leq 0$: take $n = 1 - 2k$ (a positive odd integer). Then $f(n) = -(n-1)/2 = -(-2k)/2 = k$. ✓

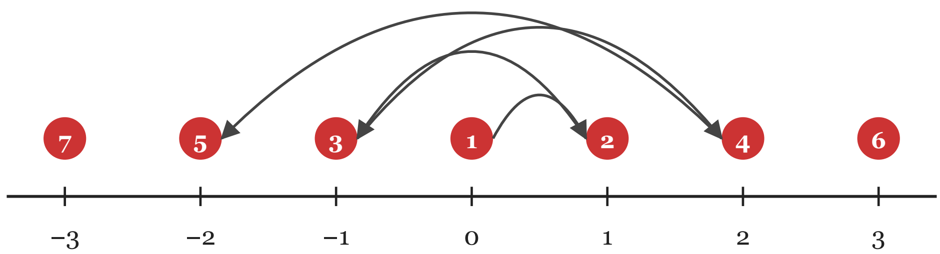
So f hits every integer.

Injective. Even n map to *positive* integers and odd n map to *non-positive* integers — disjoint outputs by parity. Within each parity class, f is strictly monotone in n, hence one-to-one.

So f is a bijection — the listing visits every integer exactly once. ■

1.5 Visual — The Zigzag Path

Reading f off the number line: start at 0 and alternate right-positive and left-negative, growing in magnitude by one each cycle.



...path continues to position 6 (= integer 3), then 7 (= −3), and so on

Figure 1.1: Each integer carries a red position-number showing when it is visited by the listing. Arrows show the path of visits: $0 \to 1 \to -1 \to 2 \to -2 \to 3 \to -3 \to \cdots$, with the magnitude of each step growing by one.

1.6 Remark — Why the "Naïve" Listing Doesn't Count

A first instinct might be to list all non-negative integers, then all negatives:

$$\{0,\ 1,\ 2,\ 3,\ 4,\ \ldots,\ -1,\ -2,\ -3,\ -4,\ \ldots\}.$$

But this is *not* a singly infinite listing — it contains *two* ellipses, both running to infinity. At what position does -1 appear? You can't write down any finite n: the non-negatives $0, 1, 2, 3, \ldots$ alone already exhaust positions $1, 2, 3, \ldots$, leaving no room for the negatives.

This isn't just an aesthetic complaint. The ordering above has *order type* $\omega + \omega$ — two infinite "rounds" stacked. A singly infinite listing must have order type ω — a single infinite "round". You can't slip the negatives in *after* the positives because there is no "after" in ω.

The zigzag listing $0, 1, -1, 2, -2, \ldots$ has order type ω precisely because it *interleaves* the positives and negatives so that every integer gets a finite position.

1.7 Remark — Other Valid Orderings

The zigzag is convenient, but not unique. Any *bijection* $\mathbb{N} \to \mathbb{Z}$ yields a valid singly infinite listing. For instance:

- **Two-up-one-down**: $\{0, 1, 2, -1, 3, 4, -2, 5, 6, -3, \ldots\}$ — list two non-negatives, then one negative, repeating.
- **Lexicographic by magnitude, ties broken negatives first**: $\{0, -1, 1, -2, 2, -3, 3, \ldots\}$ — same as the standard zigzag with the sign convention flipped after 0.
- **Spread by prime exponents**: send $2^a 3^b \mapsto a - b$ for the obvious bijection $\mathbb{N} \to \mathbb{Z}^2$ -style construction (overkill but valid).

What makes the textbook's expected answer the *standard* one is symmetry and minimality: it is the listing in which each integer appears at position $\approx 2|k|$ — i.e. roughly at the position equal to twice its distance from 0.

1.8 Remark — This is the First Hint of Countability

The exercise is a sleight of hand: writing $\mathbb{Z}$ as a singly infinite listing **is exactly the same** as exhibiting a bijection $\mathbb{N} \to \mathbb{Z}$.

Together these say $\mathbb{Z}$ *is countable* — i.e. it has the same cardinality as $\mathbb{N}$. In Chapter 8 the textbook will develop this notion formally and prove that even $\mathbb{Q}$ (the rationals) is countable, while $\mathbb{R}$ is not (Cantor's diagonal argument).

The naïve intuition that "there are twice as many integers as natural numbers" (positives, negatives, plus zero) is *wrong* in the technical sense of cardinality. The zigzag listing is the constructive witness against that intuition: it pairs up the integers with the naturals one-for-one, with no integer left behind.

1.9 Remark — Inverting the Listing

It's useful to be able to *read* the listing in either direction. The inverse map $g : \mathbb{Z} \to \mathbb{N}$ takes an integer k to its position in the list:

$$g(k) \;=\; \begin{cases} 2k & \text{if } k > 0, \\ 1 - 2k & \text{if } k \le 0. \end{cases}$$

So for example $g(0) = 1$, $g(1) = 2$, $g(-1) = 3$, $g(5) = 10$, $g(-5) = 11$. *Positive integers occupy the even positions; non-positive integers occupy the odd positions* — exactly the parity structure of the original f.

1.10 Remark — A Compact Pictorial Mnemonic

A one-line way to remember the zigzag is the spiral

$$0 \;\to\; +1 \;\to\; -1 \;\to\; +2 \;\to\; -2 \;\to\; +3 \;\to\; -3 \;\to\; \cdots$$

or pictorially:

$$0 \nearrow 1 \swarrow -1 \nearrow 2 \swarrow -2 \nearrow 3 \swarrow \cdots$$

Both are descriptions of the same listing — alternating up-and-down across the integer line, starting at 0.